Vectors I Test — Solutions

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Full sourced questions. Give exact answers unless otherwise stated.

1 [DHS 2023 Promo Q5]

With reference to the origin O, the points A and B have coordinates $(1, 2, 0)$ and $(-5, 1, 0)$.

(a) Find the angle AOB. [3]

The vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are projected along the direction of $\begin{pmatrix}1\\2\\2\end{pmatrix}$ and $\begin{pmatrix}2\\1\\2\end{pmatrix}$ to form the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$ respectively.

(b) Find the vectors $\overrightarrow{O'A'}$ and $\overrightarrow{O'B'}$. [3]

(c) Find the exact area of the triangle $O'A'B'$. [2]

Solution

(a) $\overrightarrow{OA} = \begin{pmatrix}1\\2\\0\end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix}-5\\1\\0\end{pmatrix}$. Let $\theta = \angle AOB$.

$$\overrightarrow{OA} \cdot \overrightarrow{OB} = |\overrightarrow{OA}||\overrightarrow{OB}| \cos\theta$$

$$\cos\theta = \frac{\begin{pmatrix}1\\2\\0\end{pmatrix} \cdot \begin{pmatrix}-5\\1\\0\end{pmatrix}}{\sqrt{1^2+2^2}\sqrt{5^2+1^2}} = \frac{-5+2}{\sqrt{5}\sqrt{26}} = -\frac{3}{\sqrt{5}\sqrt{26}}$$

$$\theta = 105.3°.$$

(b)

$$\overrightarrow{O'A'} = \frac{\overrightarrow{OA} \cdot \begin{pmatrix}1\\2\\2\end{pmatrix}}{\left|\begin{pmatrix}1\\2\\2\end{pmatrix}\right|} \frac{\begin{pmatrix}1\\2\\2\end{pmatrix}}{\left|\begin{pmatrix}1\\2\\2\end{pmatrix}\right|} = \frac{\begin{pmatrix}1\\2\\0\end{pmatrix} \cdot \begin{pmatrix}1\\2\\2\end{pmatrix}}{(\sqrt{1^2+2^2+2^2})^2} \begin{pmatrix}1\\2\\2\end{pmatrix} = \frac{5}{9}\begin{pmatrix}1\\2\\2\end{pmatrix}$$

$$\overrightarrow{O'B'} = \frac{\overrightarrow{OB} \cdot \begin{pmatrix}2\\1\\2\end{pmatrix}}{\left|\begin{pmatrix}2\\1\\2\end{pmatrix}\right|} \frac{\begin{pmatrix}2\\1\\2\end{pmatrix}}{\left|\begin{pmatrix}2\\1\\2\end{pmatrix}\right|} = \frac{\begin{pmatrix}-5\\1\\0\end{pmatrix} \cdot \begin{pmatrix}2\\1\\2\end{pmatrix}}{(\sqrt{2^2+1^2+2^2})^2} \begin{pmatrix}2\\1\\2\end{pmatrix} = -\begin{pmatrix}2\\1\\2\end{pmatrix}.$$

(c)

$$\text{Area of triangle } O'A'B' = \frac{1}{2}|\overrightarrow{O'A'} \times \overrightarrow{O'B'}|$$

$$= \frac{5}{18}\left|\begin{pmatrix}1\\2\\2\end{pmatrix} \times \begin{pmatrix}2\\1\\2\end{pmatrix}\right| \qquad (|\mathbf{a} \times (-\mathbf{b})| = |\mathbf{a} \times \mathbf{b}|)$$

$$= \frac{5}{18}\left|\begin{pmatrix}2\\2\\-3\end{pmatrix}\right|$$

$$= \frac{5}{18}\sqrt{17} \text{ units}^2.$$

2 [NYJC 2024 Promo Q5]

The position vectors of points A, B and C, relative to the origin O are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a}=3\mathbf{i}+2\mathbf{j}-\mathbf{k}, \quad \mathbf{b}=4\mathbf{i}-3\mathbf{j}+2\mathbf{k}, \quad \mathbf{c}=3\mathbf{i}-\mathbf{j}-\mathbf{k}.$$

(a) Find $\mathbf{a}\times\mathbf{b}$ and deduce the exact area of triangle OAB. [3]

(b) Hence, find the volume of the tetrahedron $OABC$.

[The volume of a tetrahedron is $\frac{1}{3}\times$base area$\times$perpendicular height.] [3]

(c) Find the perpendicular distance from B to line OA. [2]

Solution

(a)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}3\\2\\-1\end{pmatrix}\times\begin{pmatrix}4\\-3\\2\end{pmatrix}=\begin{pmatrix}1\\-10\\-17\end{pmatrix}$$

$$\begin{aligned}\text{Area of } \triangle AOB &= \frac{1}{2}|\mathbf{a}\times\mathbf{b}| \\ &= \frac{1}{2}\left|\begin{pmatrix}1\\-10\\-17\end{pmatrix}\right| \\ &= \frac{1}{2}\sqrt{1+100+289}=\frac{1}{2}\sqrt{390}\text{ units}^2.\end{aligned}$$

(b)

$$\begin{aligned}\text{Perpendicular height of tetrahedron} &= \frac{1}{\sqrt{1+100+289}}\left|\overrightarrow{OC}\cdot\begin{pmatrix}1\\-10\\-17\end{pmatrix}\right| \\ &= \frac{1}{\sqrt{390}}\left|\begin{pmatrix}3\\-1\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\-10\\-17\end{pmatrix}\right| \\ &= \frac{30}{\sqrt{390}}\end{aligned}$$

$$\text{Volume of tetrahedron } OABC=\frac{1}{3}\left(\frac{\sqrt{390}}{2}\right)\left(\frac{30}{\sqrt{390}}\right)=5\text{ units}^3.$$

(c)

$$\begin{aligned}\text{Perpendicular distance from } B \text{ to line } OA &= |\mathbf{b}\times\hat{\mathbf{a}}| \\ &= \frac{\sqrt{390}}{\sqrt{9+4+1}} \\ &= \sqrt{\frac{195}{7}} \quad \text{or } 5.28.\end{aligned}$$

3 [MI 2024 Promo Q4]

Referred to the origin O, the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. C lies on AB such that $AC:AB=1:3$.

(i) Find $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [2]

(ii) Show that the area of triangle OAC is given by $k|\mathbf{a}\times\mathbf{b}|$, where k is a constant to be determined. [2]

It is now given that $\mathbf{a}$ and $\mathbf{b}$ are unit vectors.

(iii) Using the result in part (i), find the value of $(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c}$. [3]

Solution

(i)

$$AC:AB=1:3 \quad \Rightarrow \quad AC:CB=1:2$$
$$\mathbf{c}=\frac{(1)\mathbf{b}+(2)\mathbf{a}}{1+2}=\frac{2\mathbf{a}+\mathbf{b}}{3}.$$

(ii)

$$\begin{aligned}\text{Area of triangle } OAC &= \frac{1}{3}\text{ Area of triangle } OAB\\ &=\frac{1}{3}\times\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|\\ &=\frac{1}{6}|\mathbf{a}\times\mathbf{b}|, \qquad k=\frac{1}{6}.\end{aligned}$$

(iii)

$$\begin{aligned}(2\mathbf{a}-\mathbf{b})\cdot\mathbf{c} &= (2\mathbf{a}-\mathbf{b})\cdot\left(\frac{2\mathbf{a}+\mathbf{b}}{3}\right)\\ &=\frac{1}{3}(2\mathbf{a}-\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})\\ &=\frac{1}{3}(4\mathbf{a}\cdot\mathbf{a}+2\mathbf{a}\cdot\mathbf{b}-2\mathbf{b}\cdot\mathbf{a}-\mathbf{b}\cdot\mathbf{b})\\ &=\frac{1}{3}(4|\mathbf{a}|^2-|\mathbf{b}|^2)\\ &=\frac{1}{3}\left(4(1)^2-(1)^2\right) \quad \text{since } \mathbf{a}, \mathbf{b} \text{ are unit vectors}\\ &=1.\end{aligned}$$

4 [TMJC 2024 Promo Q6]

(a) The points A, B and C have position vectors $\begin{pmatrix}1\\3\\-7\end{pmatrix}$, $\begin{pmatrix}5\\1\\1\end{pmatrix}$ and $\begin{pmatrix}-9\\8\\-27\end{pmatrix}$, with respect to the origin O.

(i) Show that the points A, B and C are collinear. [2]

(ii) Find the exact area of the triangle OAB. [3]

(b) The vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are defined such that $|\mathbf{u}|=1$, $|\mathbf{v}|=2$ and $|\mathbf{w}|=3$.

(i) Suppose that $\mathbf{u}+\mathbf{v}=\mathbf{w}$. With the aid of a diagram, or otherwise, explain why $\mathbf{u}$ is a unit vector that is parallel to $\mathbf{w}$. [2]

(ii) Given instead that each of the three vectors is perpendicular to the sum of the other two vectors, find the exact value of $|\mathbf{u}+\mathbf{v}+\mathbf{w}|$. [4]

Solution

(a)(i)

$$\overrightarrow{AB}=\begin{pmatrix}5\\1\\1\end{pmatrix}-\begin{pmatrix}1\\3\\-7\end{pmatrix}=\begin{pmatrix}4\\-2\\8\end{pmatrix}$$

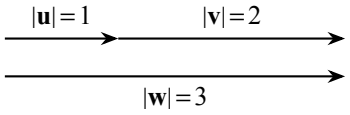
$$\overrightarrow{AC}=\begin{pmatrix}-9\\8\\-27\end{pmatrix}-\begin{pmatrix}1\\3\\-7\end{pmatrix}=\begin{pmatrix}-10\\5\\-20\end{pmatrix}=-\frac{5}{2}\begin{pmatrix}4\\-2\\8\end{pmatrix}=-\frac{5}{2}\overrightarrow{AB}.$$

Since $\overrightarrow{AB}$ is parallel to $\overrightarrow{AC}$, with a common point A, the points A, B and C are collinear. (Shown)

(a)(ii)

$$\begin{aligned}\text{Area of triangle } OAB&=\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|\\&=\frac{1}{2}\left|\begin{pmatrix}1\\3\\-7\end{pmatrix}\times\begin{pmatrix}5\\1\\1\end{pmatrix}\right|\\&=\frac{1}{2}\left|\begin{pmatrix}10\\-36\\-14\end{pmatrix}\right|=\sqrt{398}.\end{aligned}$$

(b)(i) Since $\mathbf{u}+\mathbf{v}=\mathbf{w}$, and $|\mathbf{u}|=1$, $|\mathbf{v}|=2$ and $|\mathbf{w}|=3$, $\mathbf{u}$ and $\mathbf{v}$ must both be parallel to $\mathbf{w}$, as seen in the diagram.



Since $|\mathbf{u}|=1$, $\mathbf{u}$ is a unit vector parallel to $\mathbf{w}$.

(b)(ii) Since each of the three vectors is perpendicular to the sum of the other two vectors,

$$\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=0,\qquad \mathbf{v}\cdot(\mathbf{u}+\mathbf{w})=0,\qquad \mathbf{w}\cdot(\mathbf{v}+\mathbf{u})=0$$

$$\begin{aligned}|\mathbf{u}+\mathbf{v}+\mathbf{w}|^2&=(\mathbf{u}+\mathbf{v}+\mathbf{w})\cdot(\mathbf{u}+\mathbf{v}+\mathbf{w})\\&=\mathbf{u}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v}+\mathbf{w}\cdot\mathbf{w}+\mathbf{u}\cdot(\mathbf{v}+\mathbf{w})+\mathbf{v}\cdot(\mathbf{u}+\mathbf{w})+\mathbf{w}\cdot(\mathbf{v}+\mathbf{u})\\&=|\mathbf{u}|^2+|\mathbf{v}|^2+|\mathbf{w}|^2\\&=1^2+2^2+3^2=14.\end{aligned}$$

Since $|\mathbf{u}+\mathbf{v}+\mathbf{w}|>0$, $|\mathbf{u}+\mathbf{v}+\mathbf{w}|=\sqrt{14}$.

5 [RI 2025 Prelim P1 Q3]

Relative to the origin O, the points A and C have position vectors $\mathbf{a}$ and $2\mathbf{b}$ such that

$$\mathbf{a}=5p\mathbf{i}-2p\mathbf{j}+4p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k},$$

where p is a positive constant. It is given that $|\mathbf{a}|=2|\mathbf{b}|$.

(a) Find the exact value of p. [2]

(b) Evaluate $(\mathbf{a}+2\mathbf{b})\cdot(\mathbf{a}-2\mathbf{b})$. [1]

(c) Evaluate $\mathbf{a}\times\mathbf{b}$ and hence find the area of triangle OAC. [3]

(d) Use a geometrical reason to explain why $|\mathbf{a}\times\mathbf{b}|=\frac{1}{4}|\mathbf{a}+2\mathbf{b}||\mathbf{a}-2\mathbf{b}|$. [2]

Solution

(a) Since $|\mathbf{a}|=2|\mathbf{b}|$,

$$\begin{aligned}\sqrt{(5p)^2+(-2p)^2+(4p)^2}&=2\sqrt{1^2+(-2)^2+2^2}\\ 3\sqrt{5}p&=6 \qquad (p>0)\\ p&=\frac{2}{\sqrt{5}}=\frac{2\sqrt{5}}{5}.\end{aligned}$$

(b)

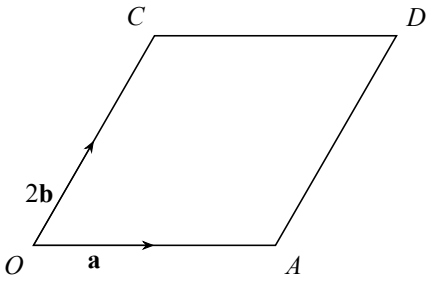
$$\begin{aligned}(\mathbf{a}+2\mathbf{b})\cdot(\mathbf{a}-2\mathbf{b})&=\mathbf{a}\cdot\mathbf{a}-2\mathbf{a}\cdot\mathbf{b}+2\mathbf{b}\cdot\mathbf{a}-4\mathbf{b}\cdot\mathbf{b}\\ &=|\mathbf{a}|^2-4|\mathbf{b}|^2 \qquad (|\mathbf{a}|=2|\mathbf{b}|)\\ &=0.\end{aligned}$$

(c)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}5p\\-2p\\4p\end{pmatrix}\times\begin{pmatrix}1\\-2\\2\end{pmatrix}=p\begin{pmatrix}5\\-2\\4\end{pmatrix}\times\begin{pmatrix}1\\-2\\2\end{pmatrix}=p\begin{pmatrix}4\\-6\\-8\end{pmatrix}=\frac{4\sqrt{5}}{5}\begin{pmatrix}2\\-3\\-4\end{pmatrix}$$

$$\begin{aligned}\text{Area of triangle } OAC&=\frac{1}{2}|\mathbf{a}\times2\mathbf{b}|=|\mathbf{a}\times\mathbf{b}|\\ &=\frac{4\sqrt{5}}{5}\sqrt{2^2+(-3)^2+(-4)^2}=\frac{4\sqrt{145}}{5}.\end{aligned}$$

(d) Observe that $\overrightarrow{OD}=\mathbf{a}+2\mathbf{b}$ and $\overrightarrow{CA}=\mathbf{a}-2\mathbf{b}$. From (b), OD is perpendicular to CA. In addition, CA and OD, which are the diagonals of the parallelogram $OADC$, bisect each other.



From (c), area of triangle $OAC=|\mathbf{a}\times\mathbf{b}|$. But area of triangle OAC is also

$$\frac{1}{2}(CA)\left(\frac{1}{2}OD\right)=\frac{1}{4}|\mathbf{a}-2\mathbf{b}||\mathbf{a}+2\mathbf{b}|.$$

Hence $|\mathbf{a}\times\mathbf{b}|=\frac{1}{4}|\mathbf{a}+2\mathbf{b}||\mathbf{a}-2\mathbf{b}|$.

END OF TEST